

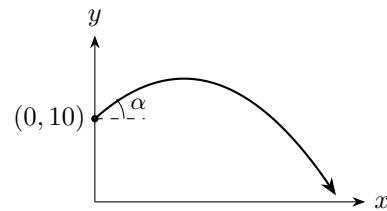
Week 4 Tutorial Solutions

Problem 15

Consider a ball that is being launched from the point $(0, 10)$, towards the right, at an angle α from horizontal. The ball will have an initial speed of 12 m/s. It will be subject to the force of gravity, but we will ignore air resistance.

- (a) From intuition, what launch angle should lead to the ball going furthest?
- (b) Starting with the acceleration vector, find the position of the ball over time, $\mathbf{r}(t)$.
- (c) Find the horizontal distance traveled by the ball if it is launched at 45 degrees, and at 30 degrees. The ground is flat, at height $y = 0$.
- (d) Comment on your initial answer in part (a).

- Projectile motion
- Launched from $(0, 10)$
- Initial speed 12 m/s at angle α



(a)

Intuitively, launching the ball at 45° will maximize the ball's horizontal range.

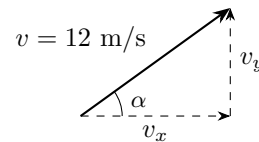
(b)

The acceleration is due to gravity only (disregard air resistance). Gravity, 9.8 m/s^2 , gives

$$\mathbf{a}(t) = \langle 0, -9.8 \rangle$$

Initial velocity components:

$$v_{0x} = 12 \cos(\alpha) \quad v_{0y} = 12 \sin(\alpha)$$



Note

We can also use the antiderivatives method used in #18.

Kinematics equations (constant acceleration, finding position):

$$\begin{aligned} x: \quad x(t) &= x_0 + v_{0x}t + \frac{1}{2}a_x t^2 & (x_0 = 0, a_x = 0) \\ x(t) &= 12t \cos(\alpha) \end{aligned} \tag{1}$$

$$\begin{aligned} y: \quad y(t) &= y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \\ &= 10 + 12t \sin(\alpha) + \frac{1}{2}(-9.8)t^2 \\ y(t) &= 10 + 12t \sin(\alpha) - 4.9t^2 \end{aligned} \tag{2}$$

Combining (1) and (2):

$$\mathbf{r}(t) = \langle 12t \cos(\alpha), 10 + 12t \sin(\alpha) - 4.9t^2 \rangle$$

(c)

The ball hits the ground at $y(t) = 0$. Setting (2) equal to zero:

$$y(t) = 0 = 10 + 12t \sin(\alpha) - 4.9t^2$$

(i) $\alpha = 45^\circ$

$$\sin(45^\circ) = \frac{\sqrt{2}}{2}, \quad \cos(45^\circ) = \frac{\sqrt{2}}{2}$$

$$\begin{aligned} 0 &= 10 + 12t \sin(45^\circ) - 4.9t^2 \\ &= 10 + 6\sqrt{2}t - 4.9t^2 \end{aligned}$$

$$t = 2.536 \text{ s} \quad t = -0.805 \text{ s}$$

(ii) $\alpha = 30^\circ$

$$\sin(30^\circ) = \frac{1}{2}, \quad \cos(30^\circ) = \frac{\sqrt{3}}{2}$$

$$\begin{aligned} 0 &= 10 + 12t \sin(30^\circ) - 4.9t^2 \\ &= 10 + 6t - 4.9t^2 \end{aligned}$$

$$t = 2.166 \text{ s} \quad t = -0.942 \text{ s}$$

We need $t > 0$ (the ball lands after it is launched), so we keep the positive roots. Substitute them into (1):

$$\begin{aligned} x(2.536) &= 12(2.536) \cos(45^\circ) \\ &= 6(2.536)\sqrt{2} \\ &= 21.52 \text{ m} \end{aligned}$$

$$\begin{aligned} x(2.166) &= 12(2.166) \cos(30^\circ) \\ &= 6(2.166)\sqrt{3} \\ &= 22.51 \text{ m} \end{aligned}$$

$$\therefore 30^\circ \text{ goes farther than } 45^\circ$$

(d)

The 45° intuition does not apply because the projectile does not land at the same height it is launched from. Starting from 10 m gives additional fall time.

- Height is doing the work for v_y .
- Optimal angle $< 45^\circ$.

Problem 16

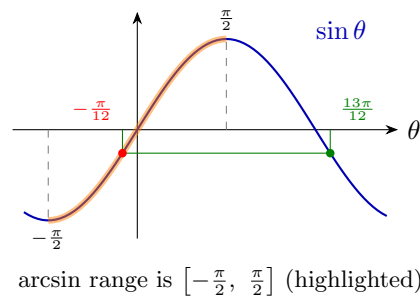
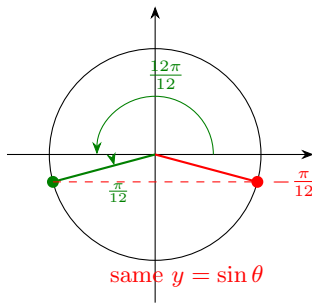
Find the exact value of the following expressions, without using a calculator.

- (a) $\arcsin\left(\sin\left(\frac{13\pi}{12}\right)\right)$
- (b) $x = \arccos\left(\cos\left(\frac{-5\pi}{8}\right)\right)$
- (c) $\cos\left(\arcsin\left(\frac{1}{2}\right)\right)$
- (d) $y = \sin\left(\arctan\left(\frac{7}{2}\right)\right)$
- (e) $\sin(\arctan(x))$

(a)

$$\arcsin\left(\sin\left(\frac{13\pi}{12}\right)\right) = \boxed{-\frac{\pi}{12}}$$

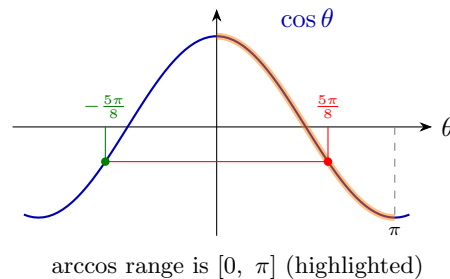
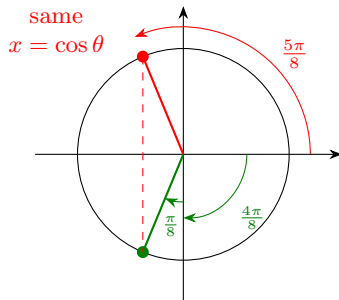
The range of arcsin is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so the answer must be an angle in that interval with the same $y = \sin \theta$ value on the unit circle.



(b)

$$x = \arccos\left(\cos\left(\frac{-5\pi}{8}\right)\right) = \boxed{\frac{5\pi}{8}}$$

The range of arccos is $[0, \pi]$, so the answer must be an angle in that interval with the same $x = \cos \theta$ value on the unit circle.

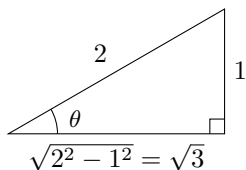


(c)

$$\cos\left(\underbrace{\arcsin\left(\frac{1}{2}\right)}_{\text{let } =\theta}\right)$$

Let θ be the angle in the triangle below:

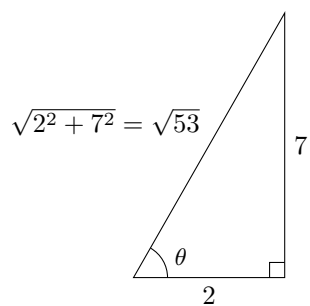
$$\Rightarrow \sin \theta = \frac{1}{2} = \frac{\text{opp}}{\text{hyp}}$$



$$\cos\left(\arcsin\left(\frac{1}{2}\right)\right) = \boxed{\frac{\sqrt{3}}{2}}$$

(d)

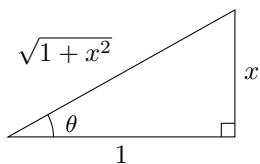
$$y = \sin\left(\underbrace{\arctan\left(\frac{7}{2}\right)}_{\text{let } =\theta}\right), \quad \text{so} \quad \tan \theta = \frac{7}{2} = \frac{\text{opp}}{\text{adj}}$$



$$\begin{aligned} y &= \sin\left(\arctan\left(\frac{7}{2}\right)\right) \\ &= \sin(\theta) \\ &= \boxed{\frac{7}{\sqrt{53}}} \end{aligned}$$

(e)

$$\sin\left(\underbrace{\arctan(x)}_{\text{let } =\theta}\right) \implies \tan(\theta) = \frac{x}{1} = \frac{\text{opp}}{\text{adj}}$$



$$\begin{aligned} \sin(\arctan(x)) &= \sin(\theta) \\ &= \boxed{\frac{x}{\sqrt{1 + x^2}}} \end{aligned}$$

Problem 17

Differentiate the following scalar and vector-valued functions.

(a) $f(s) = \frac{\arctan(s)}{4^s}$

(b) $\mathbf{r}(t) = \left\langle \frac{\arccos(t)}{e^{2t}}, 10^{\sqrt[3]{t}} \right\rangle$

Recall

$$\frac{d}{ds} \arctan(s) = \frac{1}{1+s^2} \quad \frac{d}{ds} e^s = e^s \quad \frac{d}{ds} 4^s = 4^s \ln(4)$$

(a)

Using the quotient rule:

$$f'(s) = \frac{\left(\frac{1}{1+s^2} \right) (4^s) - \arctan(s) \cdot 4^s \ln(4)}{(4^s)^2}$$

(b)

$$\mathbf{r}(t) = \left\langle \frac{\arccos(t)}{e^{2t}}, 10^{\sqrt[3]{t}} \right\rangle$$

or rewrite,

$$\mathbf{r}(t) = \left\langle \arccos(t) \cdot e^{-2t}, 10^{(t^{1/3})} \right\rangle$$

Using the quotient rule on the first form:

$$\mathbf{r}'(t) = \left\langle \frac{\frac{-1}{\sqrt{1-t^2}} e^{2t} - 2 \arccos(t) e^{2t}}{(e^{2t})^2}, \frac{1}{3} \ln(10) t^{-2/3} 10^{t^{1/3}} \right\rangle$$

Using the product rule on the rewritten form:

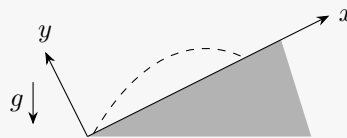
$$\mathbf{r}'(t) = \left\langle \frac{-e^{-2t}}{\sqrt{1-t^2}} - 2 \arccos(t) e^{-2t}, \frac{1}{3} \ln(10) t^{-2/3} 10^{t^{1/3}} \right\rangle$$

Both/either answer is correct.

Problem 18

You throw a ball up the side of a hill. Relative to the xy coordinate axes shown in the diagram to the right, the velocity of the ball at time t (in seconds) is given by

$$\mathbf{v}(t) = \langle \underbrace{-9t + 10}_{v_x(t)}, \underbrace{-9.8t + 6}_{v_y(t)} \rangle \text{ m/s.}$$



- Give a function $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ which describes the position of the ball at time t . Assume that the ball is thrown from $\langle x, y \rangle = \langle 0, 0 \rangle$.
- When does the ball hit the hillside?
- Since there is no wind, the only acceleration on the ball is due to gravity. Give a vector in the hillside xy coordinates which points in the direction of gravity.
- The accelerations due to gravity on Earth, Mars and Neptune are 9.8 m/s^2 , 3.8 m/s^2 and 13.3 m/s^2 respectively. Which planet are we on? Be sure to justify your answer.

Velocity: $\mathbf{v}(t) = \langle -9t + 10, -9.8t + 6 \rangle \text{ m/s}$, so the x - and y -equations are:

$$x: \quad v_x(t) = 10 - 9t \quad (3)$$

$$y: \quad v_y(t) = 6 - 9.8t \quad (4)$$

(a)

$$(x_0, y_0) = (0, 0) \quad \mathbf{r}'(t) = \mathbf{v}(t)$$

Note

We could also use the same kinematics approach taken in #15.

x : What function has the derivative $10 - 9t$ (3)?

y : What function has the derivative $6 - 9.8t$ (4)?

$$\frac{d}{dt} \left(-\frac{9}{2}t^2 + 10t \right) = -9t + 10$$

$$\frac{d}{dt} \left(-\frac{9.8}{2}t^2 + 6t \right) = -9.8t + 6$$

$$x(t) = -4.5t^2 + 10t + C_1 \quad (5)$$

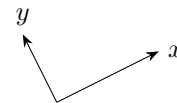
$$y(t) = -4.9t^2 + 6t + C_2 \quad (6)$$

Since $\mathbf{r}(0) = (0, 0)$, we get $C_1 = C_2 = 0$ (we do not need to adjust the position vector for the starting condition).

$$\mathbf{r}(t) = \langle -4.5t^2 + 10t, -4.9t^2 + 6t \rangle$$

(b)

The x -axis is along the hillside, so the ball lands when $y(t) = 0$.



Set the y -equation (6) (with $C_2 = 0$) equal to zero:

$$\begin{aligned} y(t) = 0 &= -4.9t^2 + 6t \\ &= t(-4.9t + 6) \end{aligned}$$

$t = 0$ is the time of the throw, so we use the other factor:

$$\begin{aligned} -4.9t + 6 &= 0 \\ t &= \frac{6}{4.9} = \boxed{1.22 \text{ s}} \end{aligned}$$

(c)

Acceleration is the derivative of velocity:

$$\mathbf{a}(t) = \mathbf{v}'(t)$$

$$\mathbf{a}(t) = \boxed{\langle -9, -9.8 \rangle}$$

(d)

Use the magnitude of our found acceleration:

$$\begin{aligned}\|\mathbf{a}\| &= \sqrt{(-9)^2 + (-9.8)^2} \\ &\approx 13.3 \text{ m/s}^2\end{aligned}$$

$$\therefore \boxed{\text{We are on Neptune.}}$$